Lab 5 – System modeling (Cont.): Sequence Diagram/ Activity Diagram

Objectives: Understanding System modeling

Tasks:

- Learn how to use a UML tool to draw sequence and activity diagrams.
- Draw sequence/ activity diagrams for each use case in your project → Document.
- Design user interfaces → Document.
- Document data dictionary according to the ERD in your project.
- **Submit your lab report including tasks, diagrams, UIs, and data dictionary.**
- Prepare for the Design document (referring to the provided templates/ samples)

Tutorials

Learn how to use UMLet to draw sequence and activity diagrams

How to create sequence diagrams?

Change the current palette to "Sequence - all in one". Add the sequence diagram element to the diagram via double-click.

This element's markup language is slightly more complex. The main idea is that each lane has a name and an ID (defined by the string "_name~ID_"). The IDs can then be used to define messages between lanes, e.g., "id1->id3".

Ex: https://www.itmeyer.at/umlet/uml2/InteractionDiagram2.html

How to create activity diagrams?

Change the current palette to "Activity - all in one". Add the activity diagram element to the diagram via double-click.
Here, TABs in the element description are used to define the activity forks.

Ex: https://www.itmeyer.at/umlet/uml2/ActivityClass1.html

More References on UMLet

[1] http://www.umlet.com/

[2] Sample diagrams in UMLet: http://www.itmeyer.at/umlet/uml2

Step by step, create a sequence diagram.

1. Identify a set of objects that will participate in the general collaboration (or use case scenario)
    a. If you derive the sequence diagram based on a scenario of a use case, select the everyday scenarios first.
    b. You should know the primary actor(s) who activates the use case.
2. Consider the first point of the scenario (or if you get it from the first point of the flow of event of a use case)
3. Consider what the system needs to do to respond to the actor when the actor sends the message to the system.
    a. What needs to be handled before the return message response from the system?
    b. Think about a sequence of events: what is input and output? Who sent the input and received the output?
    c. In this way, you will identify the candidate objects and operations of the target application for that particular scenario. You can also use this information to derive the class diagram incrementally.
4. Repeat each point of the scenario (or flow of event) until you complete all the points in the scenario.

Tutorials:

- Tutorial about Sequence Diagram:
    - https://www.visual-paradigm.com/guide/uml-unified-modeling-language/what-is-sequence-diagram/
    - https://creately.com/blog/diagrams/sequence-diagram-tutorial/
- How to draw Sequence Diagram using Visual Paradigm: https://www.visual-paradigm.com/support/documents/vpuserguide/94/2577/7025_drawingseque.html
- Some examples: https://circle.visual-paradigm.com/category/sequence-diagram/